

GAN Objective Function

GANs play a game - the **discriminator network** tries to identify real data while the **generator network** tries to create data convincing enough to fool the discriminator. Easy enough, but what about the pesky objective function? Let's break it down.

The diagram illustrates the GAN objective function with several annotations:

- min** (yellow) and **max** (blue) labels are placed above the E terms.
- G** (orange) and **D** (blue) labels are placed below the E terms.
- A red arrow points from the text "This is the distribution of real data" to the $p_{data}(x)$ term.
- A red arrow points from the text "This is the distribution of random noise fed to G" to the $p_z(z)$ term.
- A purple arrow points from the text "The purple expression shows the probability that D will classify real data correctly" to the $D(x)$ term.
- A green arrow points from the text "The green expression shows the probability that D will recognize generated data as fake" to the $D(G(z))$ term.

The generator **G** wants to minimize the expression

The discriminator **D** wants to maximize the expression

D wants the biggest value of the expression, and G wants the smallest value!

$$\min_G \max_D E_{x \sim p_{data}(x)} [\log D(x)] + E_{z \sim p_z(z)} [\log(1 - D(G(z)))]$$